

The Elastodynamic Equation: Body Waves and Homogeneous Media Solutions

September, 2025

Elastodynamic Equation

The goal is to derive equation (1) below:

$$\boxed{\rho \ddot{\mathbf{u}} = (\lambda + 2\mu) \nabla(\nabla \cdot \mathbf{u}) - \mu \nabla \times (\nabla \times \mathbf{u})} \quad (1)$$

Starting from the momentum balance and Hooke's law. We will use index notation (Einstein summation) where helpful and also show the vector identity used.

Assumptions

- Small strains, linear elasticity (infinitesimal strain tensor).
- Homogeneous, isotropic solid with constant Lamé parameters λ, μ .
- No body forces.
- $\mathbf{u}(\mathbf{x}, t)$ is displacement; ρ density.

Step 1 — Balance of linear momentum (Cauchy)

$$\rho \ddot{u}_i = \partial_j \sigma_{ij}, \quad \text{or in vector form } \rho \ddot{\mathbf{u}} = \nabla \cdot \boldsymbol{\sigma}.$$

Step 2 — Constitutive law (isotropic Hooke's law)

Strain (infinitesimal):

$$\varepsilon_{kl} = \frac{1}{2}(\partial_k u_l + \partial_l u_k).$$

Stress:

$$\sigma_{ij} = c_{ijkl} \varepsilon_{kl} = \lambda \delta_{ij} \varepsilon_{kk} + 2\mu \varepsilon_{ij}.$$

Since $\varepsilon_{kk} = \partial_k u_k$ and $\varepsilon_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$, we get

$$\boxed{\sigma_{ij} = \lambda \delta_{ij} (\partial_k u_k) + \mu (\partial_i u_j + \partial_j u_i) .} \quad (2)$$

Step 3 — Compute $\partial_j \sigma_{ij}$

Differentiate (1) with respect to x_j :

$$\partial_j \sigma_{ij} = \partial_j (\lambda \delta_{ij} \partial_k u_k) + \partial_j (\mu (\partial_i u_j + \partial_j u_i)).$$

With constant λ, μ (homogeneous medium) and δ_{ij} constant:

$$\partial_j (\lambda \delta_{ij} \partial_k u_k) = \lambda \partial_i (\partial_k u_k).$$

For the second term:

$$\partial_j (\mu (\partial_i u_j + \partial_j u_i)) = \mu (\partial_j \partial_i u_j + \partial_j \partial_j u_i).$$

So

$$\boxed{\partial_j \sigma_{ij} = \lambda \partial_i (\partial_k u_k) + \mu \partial_j \partial_i u_j + \mu \partial_j \partial_j u_i .} \quad (3)$$

Step 4 — Use commutativity of partial derivatives

Since mixed partials commute for smooth fields,

$$\partial_j \partial_i u_j = \partial_i (\partial_j u_j).$$

Define $\nabla \cdot \mathbf{u} = \partial_j u_j$ and Laplacian $\nabla^2 u_i = \partial_j \partial_j u_i$. Then (3) becomes

$$\partial_j \sigma_{ij} = \lambda \partial_i (\nabla \cdot \mathbf{u}) + \mu \partial_i (\nabla \cdot \mathbf{u}) + \mu \nabla^2 u_i$$

$$\partial_j \sigma_{ij} = (\lambda + \mu) \partial_i (\nabla \cdot \mathbf{u}) + \mu \nabla^2 u_i.$$

So the momentum equation is

$$\rho \ddot{u}_i = (\lambda + \mu) \partial_i (\nabla \cdot \mathbf{u}) + \mu \nabla^2 u_i. \quad (4a)$$

In vector form:

$$\boxed{\rho \ddot{\mathbf{u}} = (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u}} \quad (4b)$$

Step 5 — Replace $\nabla^2 \mathbf{u}$ using a vector identity

Use the standard identity

$$\boxed{\nabla^2 \mathbf{u} = \nabla (\nabla \cdot \mathbf{u}) - \nabla \times (\nabla \times \mathbf{u})} \quad (5)$$

(you may also prove (5) componentwise.)

Substitute (5) into (4b):

$$\rho \ddot{\mathbf{u}} = (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) + \mu \left[\nabla (\nabla \cdot \mathbf{u}) - \nabla \times (\nabla \times \mathbf{u}) \right].$$

Collect the $\nabla(\nabla \cdot \mathbf{u})$ terms:

$$\rho \ddot{\mathbf{u}} = (\lambda + 2\mu) \nabla(\nabla \cdot \mathbf{u}) - \mu \nabla \times (\nabla \times \mathbf{u}).$$

This is the desired equation (1).

$$\boxed{\rho \ddot{\mathbf{u}} = (\lambda + 2\mu) \nabla(\nabla \cdot \mathbf{u}) - \mu \nabla \times (\nabla \times \mathbf{u}) .} \quad (1)$$

Index-notation derivation (compact)

From

$$\rho \ddot{u}_i = \partial_j \sigma_{ij}$$

$$\sigma_{ij} = \lambda \delta_{ij} \partial_k u_k + \mu (\partial_i u_j + \partial_j u_i)$$

Compute $\partial_j \sigma_{ij} = \lambda \partial_i (\partial_k u_k) + \mu \partial_j \partial_j u_i + \mu \partial_j \partial_i u_j$. Rearranging and using $\partial_j \partial_i u_j = \partial_i (\partial_j u_j)$ gives

$$\boxed{\rho \ddot{u}_i = (\lambda + \mu) \partial_i (\partial_j u_j) + \mu \partial_j \partial_j u_i}$$

which is algebraically equivalent to (1) after grouping into $\nabla(\nabla \cdot \mathbf{u})$ and $\nabla^2 \mathbf{u}$ and using the identity for $\nabla^2 \mathbf{u}$.

Appendix — component proof of identity (5)

Show $(\nabla \times (\nabla \times \mathbf{u}))_i = \partial_i (\partial_j u_j) - \nabla^2 u_i$.

Start with Levi-Civita:

$$(\nabla \times (\nabla \times \mathbf{u}))_i = \epsilon_{ijk} \partial_j (\epsilon_{klm} \partial_l u_m) = \epsilon_{ijk} \epsilon_{klm} \partial_j \partial_l u_m.$$

Use the epsilon–delta identity

$$\epsilon_{ijk}\epsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl},$$

so

$$(\nabla \times (\nabla \times \mathbf{u}))_i = (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) \partial_j \partial_l u_m = \partial_j \partial_i u_j - \partial_j \partial_j u_i.$$

But $\partial_j \partial_i u_j = \partial_i (\partial_j u_j) = \partial_i (\nabla \cdot \mathbf{u})$ and $\partial_j \partial_j u_i = \nabla^2 u_i$. Hence

$$\boxed{(\nabla \times (\nabla \times \mathbf{u}))_i = \partial_i (\nabla \cdot \mathbf{u}) - \nabla^2 u_i}$$

which rearranges to equation (5).

General Note

- The first term $(\lambda + 2\mu)\nabla(\nabla \cdot \mathbf{u})$ is the *irrotational* (dilatational) part $\rightarrow$ P waves.
- The second term $-\mu\nabla \times (\nabla \times \mathbf{u})$ is the *solenoidal* (shear) part $\rightarrow$ S waves.

Homogeneous Media Solutions (Plane vs Spherical Waves)

Plane Wave Solutions and Polarization

We start from the homogeneous isotropic elastodynamic equation (*already established above*):

$$\boxed{\rho \ddot{\mathbf{u}} = (\lambda + 2\mu) \nabla(\nabla \cdot \mathbf{u}) - \mu \nabla \times (\nabla \times \mathbf{u}) .} \quad (1)$$

We seek harmonic **plane-wave** solutions to the elastodynamic equation.

Step 1 — Plane-wave ansatz

Assume

$$\mathbf{u}(\mathbf{x}, t) = \mathbf{A} e^{i(k_1 x_1 + k_2 x_2 + k_3 x_3 - \omega t)} = \mathbf{A} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$$

where $\mathbf{k}$ is the wavevector, ω angular frequency, and $\mathbf{A}$ (constant) amplitude / polarization vector.

For this ansatz the differential operators act as:

$$\nabla \mapsto i\mathbf{k}, \quad \partial_t \mapsto -i\omega, \quad \nabla^2 \mapsto -|\mathbf{k}|^2.$$

So compute the pieces needed:

$$\begin{aligned} \nabla \cdot \mathbf{u} &= i(\mathbf{k} \cdot \mathbf{A}) e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} \\ \nabla(\nabla \cdot \mathbf{u}) &= -\mathbf{k}(\mathbf{k} \cdot \mathbf{A}) e^{i(\cdot)} \\ \nabla \times \mathbf{u} &= i(\mathbf{k} \times \mathbf{A}) e^{i(\cdot)} \\ \nabla \times (\nabla \times \mathbf{u}) &= -\mathbf{k} \times (\mathbf{k} \times \mathbf{A}) e^{i(\cdot)} \\ \ddot{\mathbf{u}} &= -\omega^2 \mathbf{A} e^{i(\cdot)} \end{aligned}$$

(Note that $e^{i(\cdot)}$ is short for $e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$.)

Step 2 - Substitute into equation (1) and cancel the common exponential factor

Substitute all terms into (1). After cancelling the common factor $e^{i(\cdot)}$ we get

$$\rho(-\omega^2)\mathbf{A} = (\lambda + 2\mu)[-\mathbf{k}(\mathbf{k} \cdot \mathbf{A})] - \mu[-\mathbf{k} \times (\mathbf{k} \times \mathbf{A})].$$

Simplify signs:

$$-\rho\omega^2\mathbf{A} = -(\lambda + 2\mu)\mathbf{k}(\mathbf{k} \cdot \mathbf{A}) + \mu\mathbf{k} \times (\mathbf{k} \times \mathbf{A}).$$

Multiply both sides by -1 :

$$\rho\omega^2\mathbf{A} = (\lambda + 2\mu)\mathbf{k}(\mathbf{k} \cdot \mathbf{A}) - \mu\mathbf{k} \times (\mathbf{k} \times \mathbf{A}). \tag{6}$$

Step 3 - Use the vector triple-product identity

Use

$$\mathbf{k} \times (\mathbf{k} \times \mathbf{A}) = \mathbf{k}(\mathbf{k} \cdot \mathbf{A}) - |\mathbf{k}|^2 \mathbf{A}.$$

Plugging this into (6):

$$\begin{aligned} \rho\omega^2 \mathbf{A} &= (\lambda + 2\mu) \mathbf{k}(\mathbf{k} \cdot \mathbf{A}) - \mu [\mathbf{k}(\mathbf{k} \cdot \mathbf{A}) - |\mathbf{k}|^2 \mathbf{A}] \\ &= (\lambda + 2\mu - \mu) \mathbf{k}(\mathbf{k} \cdot \mathbf{A}) + \mu |\mathbf{k}|^2 \mathbf{A} \\ &= (\lambda + \mu) \mathbf{k}(\mathbf{k} \cdot \mathbf{A}) + \mu |\mathbf{k}|^2 \mathbf{A}. \end{aligned}$$

Rearrange:

$$\boxed{(\rho\omega^2 - \mu|\mathbf{k}|^2) \mathbf{A} = (\lambda + \mu) \mathbf{k}(\mathbf{k} \cdot \mathbf{A}).} \quad (7)$$

Equation (7) is an algebraic eigenvalue-type equation for $\mathbf{A}$.

Step 4 - Dot equation (7) with $\mathbf{k}$ to find a scalar relation

Take the dot product of (7) with $\mathbf{k}$:

$$(\rho\omega^2 - \mu|\mathbf{k}|^2) (\mathbf{k} \cdot \mathbf{A}) = (\lambda + \mu) |\mathbf{k}|^2 (\mathbf{k} \cdot \mathbf{A}).$$

Move all terms to one side:

$$\boxed{(\rho\omega^2 - (\lambda + 2\mu)|\mathbf{k}|^2) (\mathbf{k} \cdot \mathbf{A}) = 0} \quad (8)$$

Thus **either**

- $\mathbf{k} \cdot \mathbf{A} = 0$ (transverse polarization), **or**
- $\mathbf{k} \cdot \mathbf{A} \neq 0$ (longitudinal polarization)
- $\rho\omega^2 = (\lambda + 2\mu)|\mathbf{k}|^2$ (longitudinal branch).

We now have two possibilities.

Case A — Longitudinal (P) wave: $\mathbf{k} \cdot \mathbf{A} \neq 0$

If $\mathbf{k} \cdot \mathbf{A} \neq 0$, equation (8) forces

$$(\rho\omega^2 - (\lambda + 2\mu)|\mathbf{k}|^2) = \frac{0}{(\mathbf{k} \cdot \mathbf{A})} \quad (9)$$

$$\rho\omega^2 = (\lambda + 2\mu) |\mathbf{k}|^2$$

Hence

$$\omega = \pm \sqrt{\frac{\lambda + 2\mu}{\rho}} |\mathbf{k}|$$

Define the compressional (P) speed

$$\boxed{\alpha = \sqrt{\frac{\lambda + 2\mu}{\rho}}, \quad \omega = \pm \alpha |\mathbf{k}|}$$

Now, let's show that $\mathbf{A}$ must be parallel to $\mathbf{k}$.

Using equation (7):

$$(\rho\omega^2 - \mu|\mathbf{k}|^2) \mathbf{A} = (\lambda + \mu) \mathbf{k}(\mathbf{k} \cdot \mathbf{A}).$$

Remember:

$$\rho\omega^2 = (\lambda + 2\mu) |\mathbf{k}|^2$$

$$((\lambda + 2\mu) |\mathbf{k}|^2 - \mu|\mathbf{k}|^2) \mathbf{A} = (\lambda + \mu) \mathbf{k}(\mathbf{k} \cdot \mathbf{A})$$

$$(\lambda |\mathbf{k}|^2 + 2\mu |\mathbf{k}|^2 - \mu |\mathbf{k}|^2) \mathbf{A} = (\lambda + \mu) \mathbf{k}(\mathbf{k} \cdot \mathbf{A})$$

$$(\lambda + \mu) |\mathbf{k}|^2 \mathbf{A} = (\lambda + \mu) \mathbf{k}(\mathbf{k} \cdot \mathbf{A})$$

Thus equation (7) becomes

$$\boxed{(\lambda + \mu) |\mathbf{k}|^2 \mathbf{A} = (\lambda + \mu) \mathbf{k}(\mathbf{k} \cdot \mathbf{A})}$$

Cancel $(\lambda + \mu)$ (nonzero for ordinary solids) and divide by $|\mathbf{k}|^2$:

$$\mathbf{A} = \frac{\mathbf{k}(\mathbf{k} \cdot \mathbf{A})}{|\mathbf{k}|^2}.$$

This means $\mathbf{A}$ equals its projection onto $\mathbf{k}$ — i.e. $\mathbf{A}$ is **parallel to $\mathbf{k}$** . Thus **P-wave polarization is longitudinal (particle motion parallel to propagation)**.

Case B — Transverse (S) wave: $\mathbf{k} \cdot \mathbf{A} = 0$

If $\mathbf{k} \cdot \mathbf{A} = 0$, equation (7) reduces to

$$(\rho\omega^2 - \mu|\mathbf{k}|^2) \mathbf{A} = \mathbf{0}.$$

Non-trivial $\mathbf{A}$ requires

$$\rho\omega^2 = \mu|\mathbf{k}|^2,$$

so

$$\omega = \pm \sqrt{\frac{\mu}{\rho}} |\mathbf{k}|.$$

Define the shear (S) speed

$$\boxed{\beta = \sqrt{\frac{\mu}{\rho}}, \quad \omega = \pm \beta |\mathbf{k}|}$$

Because $\mathbf{k} \cdot \mathbf{A} = 0$, the polarization $\mathbf{A}$ is **perpendicular to $\mathbf{k}$** (transverse). There are two

independent orthogonal polarizations (S is twofold degenerate).

Compact matrix/eigenvalue viewpoint (short form)

Rewrite (7) dividing by $|\mathbf{k}|^2$:

$$(\rho c^2) \mathbf{A} = \mu \mathbf{A} + (\lambda + \mu) \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{A}), \quad \text{with } c^2 = \frac{\omega^2}{|\mathbf{k}|^2}, \quad \hat{\mathbf{n}} = \frac{\mathbf{k}}{|\mathbf{k}|}.$$

This is an eigenproblem for the 3×3 matrix $\mu I + (\lambda + \mu) \hat{\mathbf{n}} \hat{\mathbf{n}}^T$. Its eigenvalues are:

- $\rho c^2 = \lambda + 2\mu$ for eigenvector $\hat{\mathbf{n}}$ (longitudinal),
- $\rho c^2 = \mu$ (double) for any vector orthogonal to $\hat{\mathbf{n}}$ (two shear polarizations).

Final Notes

- **P (compressional / longitudinal) waves:** $\omega = \pm \alpha |\mathbf{k}|$, $\alpha = \sqrt{\frac{\lambda+2\mu}{\rho}}$, $\mathbf{A} \parallel \mathbf{k}$.
- **S (shear / transverse) waves:** $\omega = \pm \beta |\mathbf{k}|$, $\beta = \sqrt{\frac{\mu}{\rho}}$, $\mathbf{A} \perp \mathbf{k}$.

Both are nondispersive (phase speed = group speed) in a homogeneous linear elastic medium. That is, same wave velocity in all direction (Isotropy). S-waves are absent if $\mu = 0$ (fluids).

Spherical Wave Solutions

Physical Context and Wave Equation

In this case we are dealing with waves radiating outward from a point source in a homogeneous medium. The governing equation is the standard scalar wave equation derived for the P-wave potential ϕ :

$$\frac{1}{\alpha^2} \ddot{\phi} = \nabla^2 \phi$$

Where:

- α is the wave speed (e.g., P-wave speed)
- ∇^2 is the Laplacian operator

- ϕ is the wave field (e.g., velocity potential)

The key point is that the **form of the Laplacian ∇^2 depends on the coordinate system**, which in turn dictates the nature of the solution.

In spherical coordinates (for spherically symmetric waves):

$$\nabla^2 \phi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right)$$

Part 1: First Attempt - Why Simple Plane Wave Fails

Try:

$$\phi(r, t) = A e^{i(\omega t - kr)}$$

Calculate first derivative:

$$\frac{\partial \phi}{\partial r} = -ikA e^{i(\omega t - kr)} = -ik\phi(r, t)$$

Calculate second part:

$$r^2 \frac{\partial \phi}{\partial r} = -ikr^2 \phi(r, t)$$

Apply outer derivative:

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right) = \frac{\partial}{\partial r} (-ikr^2 \phi(r, t))$$

Using product rule:

$$\begin{aligned} &= -ik \left[2r\phi(r, t) + r^2 \frac{\partial \phi(r, t)}{\partial r} \right] \\ &= -ik \left[2r\phi(r, t) + r^2 (-ik\phi(r, t)) \right] \\ &= -2ikr\phi(r, t) - k^2 r^2 \phi(r, t) \end{aligned}$$

Therefore:

$$\nabla^2 \phi = \frac{1}{r^2} [-2ikr\phi - k^2 r^2 \phi] = -\frac{2ik}{r} \phi - k^2 \phi$$

What we need for the wave equation:

$$\nabla^2 \phi = -k^2 \phi \quad (\text{where } k = \omega/\alpha)$$

There's a **Problem!**, because we have an extra term!

$$\nabla^2 \phi = -k^2 \phi - \frac{2ik}{r} \phi$$

$$\boxed{\nabla^2 \phi = - \left(k^2 - \frac{2ik}{r} \right) \phi} \quad \text{WRONG!}$$

The extra term $-2ik/r$ means this is NOT a solution. You should note that Amplitude A in the plane wave assumption above does not vary with radial distance r from the source (spherical symmetry).

Part 2: Physical Insight - Why Amplitude Must Decay

Energy conservation argument:

1. **Energy flows outward through spherical surfaces**
2. **Energy density** $\propto |\phi|^2$
3. **Surface area** of sphere $= 4\pi r^2$
4. **Total energy** through sphere must be constant

Therefore:

$$|\phi|^2 \times 4\pi r^2 = \text{constant}$$

This means:

$$|\phi| \propto \frac{1}{r}$$

The amplitude must decay as 1/r! Practically speaking, It also make sense that A should decay with r .

Part 3: Correct Solution with Amplitude Decay - The Exact 3D Spherical Wave Solution

Assuming:

$$\phi(r, t) = A(r)e^{i(\omega t - kr)}$$

where $A(r)$ is a function of r to be determined.

Find first derivative:

$$\frac{\partial \phi}{\partial r} = \frac{dA}{dr} e^{i(\omega t - kr)} - ikA e^{i(\omega t - kr)}$$

$$\frac{\partial \phi}{\partial r} = \left[\frac{dA}{dr} - ikA \right] e^{i(\omega t - kr)}$$

Multiply by r^2

$$r^2 \frac{\partial \phi}{\partial r} = r^2 \left[\frac{dA}{dr} - ikA \right] e^{i(\omega t - kr)}$$

$$r^2 \frac{\partial \phi}{\partial r} = r^2 \frac{dA}{dr} e^{i(\omega t - kr)} - ikA r^2 e^{i(\omega t - kr)}$$

Take derivative again (product rule)

$$\begin{aligned} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right) &= e^{i(\omega t - kr)} \times \\ &\left[2r \frac{dA}{dr} + r^2 \frac{d^2 A}{dr^2} - 2ikrA - ikr^2 \frac{dA}{dr} + (-ik) \left(r^2 \frac{dA}{dr} - ikr^2 A \right) \right] \end{aligned}$$

Simplifying:

$$= e^{i(\omega t - kr)} \left[r^2 \frac{d^2 A}{dr^2} + 2r \frac{dA}{dr} - 2ikr^2 \frac{dA}{dr} - 2ikrA - k^2 r^2 A \right]$$

Divide by r^2 to get $\nabla^2 \phi$

$$\nabla^2 \phi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right) = e^{i(\omega t - kr)} \left[\frac{d^2 A}{dr^2} + \frac{2}{r} \frac{dA}{dr} - 2ik \frac{dA}{dr} - \frac{2ik}{r} A - k^2 A \right]$$

Wave equation requires that:

$$\nabla^2 \phi = -\frac{\omega^2}{\alpha^2} \phi = -k^2 A e^{i(\omega t - kr)}$$

Equating both sides

$$\frac{d^2 A}{dr^2} + \frac{2}{r} \frac{dA}{dr} - 2ik \frac{dA}{dr} - \frac{2ik}{r} A - k^2 A = -k^2 A$$

The $k^2 A$ terms cancel:

$$\frac{d^2 A}{dr^2} + \frac{2}{r} \frac{dA}{dr} - 2ik \frac{dA}{dr} - \frac{2ik}{r} A = \cancel{-k^2 A} + \cancel{k^2 A}$$

$$\frac{d^2 A}{dr^2} + \frac{2}{r} \frac{dA}{dr} - 2ik \frac{dA}{dr} = \frac{2ik}{r} A \quad (\text{A})$$

This is a form of Cauchy-Euler equation.

Now, we must solve equation (A) to find $A(r)$

Rearrange the equation

$$\frac{d^2 A}{dr^2} + \left(\frac{2}{r} - 2ik \right) \frac{dA}{dr} = \frac{2ik}{r} A$$

Since we have both real and imaginary terms in (A), let's look at the **imaginary part** only.

The imaginary terms are:

$$-2ik \frac{dA}{dr} = \frac{2ik}{r} A$$

Divide both sides by $-2ik$:

$$\frac{dA}{dr} = -\frac{A}{r}$$

Solve the differential equation. This is a separable first-order ODE:

$$\frac{dA}{A} = -\frac{dr}{r}$$

Integrate both sides of the differential equation

$$\int \frac{dA}{A} = \int -\frac{dr}{r}$$

$$\ln |A| = -\ln |r| + C_1$$

$$\ln |A| = \ln \left| \frac{1}{r} \right| + C_1$$

Exponentiate

$$A = e^{\ln(1/r) + C_1} = e^{C_1} \cdot \frac{1}{r}$$

Let $C = e^{C_1}$:

$$\boxed{A(r) = \frac{C}{r}}$$

Let's verify whether this satisfies the real part

With $A = C/r$:

- $\frac{dA}{dr} = -\frac{C}{r^2}$
- $\frac{d^2A}{dr^2} = \frac{2C}{r^3}$

Substitute into the original equation (A):

$$\frac{2C}{r^3} + \frac{2}{r} \left(-\frac{C}{r^2} \right) - 2ik \left(-\frac{C}{r^2} \right) = \frac{2ik}{r} \left(\frac{C}{r} \right)$$

$$\frac{2C}{r^3} - \frac{2C}{r^3} + \frac{2ikC}{r^2} = \frac{2ikC}{r^2} \quad \checkmark$$

The equation is satisfied!

So, the final answer is:

$$\boxed{A(r) = \frac{C}{r}}$$

Taking $C = 1$ gives us:

$$A(r) = \frac{1}{r}$$

This is the **geometrical spreading factor** for spherical waves in 3D. The amplitude of the wave decays inversely with distance r because the wave's energy is being distributed over

an ever-increasing spherical surface area ($4\pi r^2$). This is a fundamental property of waves in 3D. The waveform remains unchanged (dispersionless) during propagation.

Testing for $\nabla^2\phi$

Let's verify this satisfies the wave equation. Assume $A(r) = C/r$. Then:

$$\phi = \frac{C}{r} e^{i(\omega t - kr)}$$

First derivative:

$$\frac{\partial \phi}{\partial r} = -\frac{C}{r^2} e^{i(\omega t - kr)} - ik \frac{C}{r} e^{i(\omega t - kr)} = -C \left(\frac{1}{r^2} + \frac{ik}{r} \right) e^{i(\omega t - kr)}$$

Now compute the key term for the Laplacian:

$$r^2 \frac{\partial \phi}{\partial r} = -C (1 + ikr) e^{i(\omega t - kr)}$$

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right) = -C(ik)(ikr) e^{i(\omega t - kr)} = Ck^2 r e^{i(\omega t - kr)}$$

Therefore, the Laplacian is:

$$\nabla^2 \phi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right) = \frac{1}{r^2} (Ck^2 r e^{i(\omega t - kr)}) = \frac{Ck^2}{r} e^{i(\omega t - kr)}$$

$$\boxed{\nabla^2 \phi = \frac{Ck^2}{r} e^{i(\omega t - kr)} = k^2 \frac{C}{r} e^{i(\omega t - kr)} = k^2 \phi}$$

The time derivative is:

$$\ddot{\phi} = \frac{\partial^2 \phi}{\partial t^2} = (i\omega)^2 \frac{C}{r} e^{i(\omega t - kr)} = -\omega^2 \frac{C}{r} e^{i(\omega t - kr)}$$

Plugging into the wave equation:

$$\frac{1}{\alpha^2} \ddot{\phi} = \nabla^2 \phi$$

$$\frac{1}{\alpha^2} \left(-\omega^2 \frac{C}{r} e^{i(\omega t - kr)} \right) = \frac{Ck^2}{r} e^{i(\omega t - kr)}$$

Canceling common terms, we get the dispersion relation:

$$-\frac{\omega^2}{\alpha^2} = k^2 \quad \text{or} \quad k^2 = \frac{\omega^2}{\alpha^2}$$

This confirms that k is constant, and our solution is valid. It is for this reason that solution in Part 1 above is wrong because k has an extra term which is imaginary.

Final Solution

The exact 3D spherical wave solution is:

$$\phi(r, t) = A(r) e^{i(\omega t - kr)} = \frac{1}{r} e^{i(\omega t - kr)}$$

Key Summary:

- ✓ **Geometrical spreading:** amplitude decreases as $1/r$
- ✓ **Phase propagation:** $e^{i(\omega t - kr)}$ travels at speed α
- ✓ **Energy conservation:** $|\phi|^2 \times r^2 = \text{constant}$
- ✓ **Singularity at origin:** $\phi \rightarrow \infty$ as $r \rightarrow 0$ (point source)

Part 4: The Asymptotic 2D “Spherical” (Cylindrical) Wave Solution

In many applications (e.g., a long line source), waves spread out cylindrically in a 2D plane.

The Laplacian for cylindrical coordinates with radial symmetry is:

$$\nabla^2 \phi = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right)$$

By analogy with the 3D case above, one might try a solution with a $1/\sqrt{r}$ amplitude decay, as the perimeter of a circle is $2\pi r$, suggesting intensity decays as $1/r$, and thus amplitude as $1/\sqrt{r}$.

$$\phi(r, t) = \frac{1}{\sqrt{r}} e^{i(\omega t - kr)}$$

Now, we will substitute this trial solution into the scalar wave equation to see if it works.

Computing the Derivatives

First, compute the spatial derivatives:

$$\frac{\partial \phi}{\partial r} = \left(-\frac{1}{2} r^{-3/2} - ikr^{-1/2} \right) e^{i(\omega t - kr)} = - \left(\frac{1}{2r} + ik \right) \frac{1}{\sqrt{r}} e^{i(\omega t - kr)}$$

Now compute $r \frac{\partial \phi}{\partial r}$:

$$r \frac{\partial \phi}{\partial r} = - \left(\frac{1}{2} + ikr \right) \frac{1}{\sqrt{r}} e^{i(\omega t - kr)}$$

Now compute the Laplacian term:

$$\frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) = \frac{\partial}{\partial r} \left[- \left(\frac{1}{2} + ikr \right) r^{-1/2} e^{i(\omega t - kr)} \right]$$

Let $f(r) = - \left(\frac{1}{2} + ikr \right) e^{i(\omega t - kr)}$

Using the product rule:

$$f'(r) = - \left[(ik)r^{-1/2} + \left(\frac{1}{2} + ikr \right) \left(-\frac{1}{2} r^{-3/2} - ikr^{-1/2} \right) \right] e^{i(\omega t - kr)}$$

Let's compute this carefully:

- Derivative of first part: $\frac{d}{dr} \left[\frac{1}{2} + ikr \right] = ik$
- So first term: $-(ik)r^{-1/2} e^{i(\omega t - kr)}$
- Second term: $-\left(\frac{1}{2} + ikr \right) \times \frac{d}{dr} \left[r^{-1/2} e^{i(\omega t - kr)} \right]$
- $\frac{d}{dr} \left[r^{-1/2} e^{i(\omega t - kr)} \right] = \left(-\frac{1}{2} r^{-3/2} - ikr^{-1/2} \right) e^{i(\omega t - kr)}$

Putting it all together:

$$f'(r) = -ikr^{-1/2} e^{-ikr} - \left(\frac{1}{2} + ikr \right) \left(-\frac{1}{2} r^{-3/2} - ikr^{-1/2} \right) e^{i(\omega t - kr)}$$

Multiply out the second term:

$$\begin{aligned}\left(\frac{1}{2} + ikr\right) \left(-\frac{1}{2}r^{-3/2} - ikr^{-1/2}\right) &= -\frac{1}{4}r^{-3/2} - \frac{ik}{2}r^{-1/2} - \frac{ik}{2}r^{-1/2} + k^2r^{1/2} \\ &= -\frac{1}{4}r^{-3/2} - ikr^{-1/2} + k^2r^{1/2}\end{aligned}$$

So:

$$\begin{aligned}f'(r) &= -ikr^{-1/2}e^{-ikr} - \left(-\frac{1}{4}r^{-3/2} - ikr^{-1/2} + k^2r^{1/2}\right) e^{i(\omega t - kr)} \\ &= \left(-ikr^{-1/2} + \frac{1}{4}r^{-3/2} + ikr^{-1/2} - k^2r^{1/2}\right) e^{i(\omega t - kr)} \\ &= \left(\frac{1}{4}r^{-3/2} - k^2r^{1/2}\right) e^{i(\omega t - kr)}\end{aligned}$$

Therefore:

$$\frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) = \left(\frac{1}{4}r^{-3/2} - k^2r^{1/2} \right) e^{i(\omega t - kr)}$$

Now the full Laplacian:

$$\begin{aligned}\nabla^2 \phi &= \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) = \frac{1}{r} \left(\frac{1}{4}r^{-3/2} - k^2r^{1/2} \right) e^{i(\omega t - kr)} \\ \nabla^2 \phi &= \left(\frac{1}{4r^{5/2}} - \frac{k^2}{r^{1/2}} \right) e^{i(\omega t - kr)} \quad (\text{Laplacian})\end{aligned}$$

$$\nabla^2 \phi = \left(\frac{1}{4r^2} - k^2 \right) \frac{1}{\sqrt{r}} e^{i(\omega t - kr)} = \left(\frac{1}{4r^2} - k^2 \right) \phi$$

Now compute the time derivative:

$$\ddot{\phi} = \frac{\partial^2 \phi}{\partial t^2} = -\omega^2 \frac{1}{\sqrt{r}} e^{i(\omega t - kr)} \quad (\text{Time Derivative})$$

Substituting into the Wave Equation

$$\frac{1}{\alpha^2} \ddot{\phi} = \nabla^2 \phi$$

$$\frac{1}{\alpha^2} \left(-\omega^2 \frac{1}{\sqrt{r}} e^{i(\omega t - kr)} \right) = \left(\frac{1}{4r^{5/2}} - \frac{k^2}{r^{1/2}} \right) e^{i(\omega t - kr)}$$

$$\frac{-\omega^2}{\alpha^2} \frac{1}{\sqrt{r}} e^{i(\omega t - kr)} = \left(\frac{1}{4r^2} - k^2 \right) \frac{1}{\sqrt{r}} e^{i(\omega t - kr)}$$

Multiply both sides by $\sqrt{r} e^{-i(\omega t - kr)}$:

$$-\frac{\omega^2}{\alpha^2} = \frac{1}{4r^2} - k^2$$

Rearranging:

$$\boxed{k^2 = \frac{\omega^2}{\alpha^2} + \frac{1}{4r^2}} \quad (\text{Dispersion Relation})$$

This is the key result that shows why the trial solution above fails as an **exact** solution. k^2 should be equal to $\frac{\omega^2}{\alpha^2}$.

Physical Interpretation and Asymptotic Solution

The dispersion relation $k^2 = \frac{\omega^2}{\alpha^2} + \frac{1}{4r^2}$ tells us:

1. **The wavenumber k depends on distance r** - this is unusual and problematic
2. **The wave is dispersive** - different frequency components travel at different speeds
3. **The waveform changes shape** as it propagates

However, in the **far field** ($r \gg \lambda$), the term $\frac{1}{4r^2}$ becomes negligible compared to $\frac{\omega^2}{\alpha^2}$ because:

- $k = \frac{2\pi}{\lambda}$, so $\frac{\omega^2}{\alpha^2} = k_0^2$ where k_0 is the constant wavenumber
- When $r > 10\lambda$, $\frac{1}{4r^2} \ll k_0^2$

Therefore, in the **far field**:

$$k^2 \approx \frac{\omega^2}{\alpha^2} \quad \Rightarrow \quad k \approx \frac{\omega}{\alpha} = k_0$$

Thus, in the **far field**, our trial solution becomes approximately valid:

$$\phi(r, t) \approx \frac{1}{\sqrt{r}} e^{i(\omega t - k_0 r)}$$

More generally, for any waveform:

$$\boxed{\phi(r, t) \approx \frac{1}{\sqrt{r}} f(\omega t - kr)} \quad \text{works well!}$$

So, the trial solution above is an asymptotic solution of the wave equation in the **far field** in the 2D spherical coordinate.

The $1/\sqrt{r}$ amplitude decay represents **cylindrical geometrical spreading** - the wave energy spreads over cylinders with perimeter proportional to r , so intensity decays as $1/r$, and amplitude as $1/\sqrt{r}$.

This derivation shows why 2D wave propagation is fundamentally more complex than 3D propagation, requiring the far-field approximation for simple solutions.

Summary of Part 4

Aspect	Mathematical Expression	Physical Meaning
Exact Relation	$k^2 = \frac{\omega^2}{\alpha^2} + \frac{1}{4r^2}$	Wavenumber depends on position
Far Field Condition	$r \gg \lambda$	Distance much greater than wavelength
Asymptotic Relation	$k^2 \approx \frac{\omega^2}{\alpha^2}$	Constant wavenumber recovered
Asymptotic Solution	$\phi(r, t) \approx \frac{1}{\sqrt{r}} f(\omega t - k_0 r)$	Wave with cylindrical spreading

Note: The exact 2D solution involves **Hankel functions**, but $1/\sqrt{r}$ is excellent for $r \gg \lambda$.

General Summary

Dimension	Spreading Factor	Exact Solution?	Physical Meaning
3D	$1/r$	✓ Yes	Energy over sphere area $\propto r^2$
2D	$1/\sqrt{r}$	✓ Only asymptotic	Energy over circle $\propto r$
1D	1 (no decay)	✓ Yes	Plane wave, no spreading

Wave Type	Mathematical Form	Intensity	Distance Dependence
Plane (1D)	$Ae^{i(\omega t - kx)}$	$ A ^2$	No decay
Cylindrical (2D)	$\frac{A}{\sqrt{r}}e^{i(\omega t - kr)}$ (asymptotic, not exact)	$\frac{ A ^2}{r}$	$I \propto \frac{1}{r}$
Spherical (3D)	$\frac{A}{r}e^{i(\omega t - kr)}$	$\frac{ A ^2}{r^2}$	$I \propto \frac{1}{r^2}$